◆ SNOWFLAKE · DATA ENGINEERING

Electric Vehicle Population

Washington State Department of Licensing

End-to-end data engineering solution on Snowflake

22,183 registrations	Bronze → Silver → Gold medallion	Cortex Analyst chat on your data	Insights Streamlit app

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Confidential - Oracle Restricted \Employees Only

USE CASE

This dataset shows the Battery Electric Vehicles (BEVs) and Plug-in Hybrid Electric Vehicles (PHEVs) that are currently registered through Washington State Deparment of Licensing (DOL)

Turn a raw public EV registration feed into governed, business-ready analytics and let non-technical users ask it questions in plain language.

1 Executives
YoY adoption trend · regional penetration · BEV vs PHEV mix

2 Sales & Marketing
Tesla vs field by region · CAFV incentive eligibility · trending models

3 Data & Platform
Reliable pipeline, data quality, open formats, secure sharing

THE DATA

22,183
vehicle registrations

77% / 23%
BEV vs PHEV

~45%
Tesla share

1999–2023
model years

Architecture — data flow

SOURCES

- **EV JSON file**
  Socrata export · staged
- **Postgres master data**
  OpenFlow CDC

→

BRONZE

- **EV_RAW**
  VARIANT + lineage
  append-only

→

SILVER

- ***_MASTER**
  county / utility
- **EV_REGISTRATIONS**
  Dynamic Table
  typed · deduped · flagged

→

GOLD

- **DIM_* / FACT_***
  star schema
- **AGG_* (Iceberg)**
  Parquet + Iceberg files
  S3 bucket

→

CONSUMPTION

- **Secure Share / Listing**
  consumer account
- **Cortex Analyst**
  + Streamlit App
  Insights
  + Streamlit App

Orchestration: Task DAG (ingest → refresh → DQ → notify) — spans every layer

Data Quality: DMFs · reconciliation · alerts ·outlook email ·spans every layer

TECHNOLOGY CHOICES

Technology choices and the alternatives

Every layer picks a Snowflake feature on purpose here's what we chose, what we weighed against it, and why.

Concern	Chosen	Alternatives	Deciding factor
Ingestion	**COPY INTO**	Snowpipe · Streaming	Batch file, deterministic demo
Master data	**OpenFlow CDC**	Fivetran · scripts	Native governance + CDC
Transforms	**Dynamic Tables**	dbt · Snowpark · Streams+Tasks	Declarative incremental, no orchestration code
Data quality	**DMFs + Alerts**	dbt tests · Great Expectations	Quality attached to the data
Orchestration	**Task DAG**	Airflow · dbt Cloud	Single-platform → native wins
Open format	**Iceberg on Gold**	Delta · Parquet exports	Interop pays at consumption
Sharing	**Secure share / listing**	Reader acct · extracts	Zero-copy, revocable, governed
NL analytics	**Cortex Analyst**	DIY text2SQL · BI NLQ	Verified queries = accuracy

TRADE-OFF 1

Dynamic Tables vs dbt

CHOSEN · DYNAMIC TABLES

- Declarative - you set TARGET_LAG, Snowflake handles refresh
- Incremental by default; skips when no new data
- Built-in DAG - no external scheduler
- Right for Snowflake-native SQL transforms

ALTERNATIVE · DBT

- Testing framework + docs + ecosystem
- Modular SQL, strong team workflows
- Needs an external scheduler to run
- Best when a dbt culture already exists

For this scope, DT removes orchestration code and cost-optimizes refresh itself. In a dbt shop they compose - dbt can create Dynamic Tables.

TRADE-OFF 2

Iceberg and why only on Gold

CHOSEN · ICEBERG ON GOLD

- Open format where data is consumed
- Spark / DuckDB / ML read the same files
- ACID, snapshots, schema evolution
- No lock-in on storage

ALTERNATIVE · ICEBERG EVERYWHERE

- Bronze/Silver gain interop nobody needs
- Sacrifices native-table speed + features
- Extra metadata + external volume cost
- Complexity mid-pipeline

Verdict — Bronze stays VARIANT (Time Travel), Silver native (fast incremental DT). Gold is the product other engines want so that's the only layer that goes open.

TRADE-OFF 3

Data sharing - method by need

Method	Use when	Note
Direct secure share (Chosen)	Known accounts, same region, no discoverability needed.	Simplest; zero-copy, revocable, same region, no extra storage or transfer cost
Private Listing (Alternative)	Cross-org / cross-region, discovery	Auto-fulfill replicates - adds cost
Marketplace	Public or monetized	Broad reach; governance + T&Cs
Data Exchange	Private partner hub, manage a group of accounts and offer a share to that group.	Ecosystem sharing - SI-relevant

Direct share is appropiate for a known partner, if we want to share EV Data more broadly with multiple state agencies or research institutions, covert to a private list for better governance and self-service onboarding.

BUSINESS VALUE

Business value vs a Spark / lakehouse build

Same outcome — without owning clusters, a catalog, an orchestration stack, and a sharing mechanism.

1

No cluster management

No sizing, tuning, or babysitting Spark — a warehouse **auto-suspends in 60s**

2

Per-second compute

Pay for the seconds you run; idle costs nothing XSMALL Sized.

3

Zero-copy sharing

Consumers query live data — no ETL, no copies, instant revoke.

4

Governance built-in

Policies, masking, tags, lineage native – not add-ons

5

Cortex AI without moving data

Semantic model + conversational analytics in-platform

6

One platform, one bill

Ingest → transform → serve → share → AI, governed once